

IUTO

Sport and Health: Predicting Health Conditions from Fitness and Lifestyle Data

Group Members :

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Project Summary

This project aims to study how everyday activity and lifestyle factors relate to a person's physical fitness, using a public health & fitness dataset of more than 687 000 daily records collected from 3,000 participants over the year 2024.

The Problem ? Fitness is shaped by many everyday habits at once : how often, how long and how intensely we exercise, how many steps we take, how well we sleep, our stress and hydration levels, ... but the combined effect of these factors is rarely made visible in a simple, data-driven way. Our project asks a question the data can actually answer: which lifestyle and activity factors most influence a person's fitness level, and how accurately can we predict that fitness level from their habits ?

Solution:

After exploring and cleaning the dataset, we will analyse relevant information and features (such as activity intensity, exercise duration, step counts, sleep, and stress) and then train and compare several regression models studied in class such as linear regression, decision tree, ... to predict the fitness score from lifestyle and activity indicators. Model performance will be evaluated using regression metrics (MAE, RMSE, and R²), together with an analysis of which features the models rely on most.

Expected results:

We will test whether activity-related factors (especially exercise duration and intensity) are the strongest drivers of fitness, and report honestly whether the data support this or not.

We anticipate moderate predictive performance rather than a perfect model, and we explicitly acknowledge the dataset's limitations : a single fitness score cannot capture every aspect of real-world health and one's situation (stressful work, noisy environment, ...).

Beyond the predictive task, the project will try to provide solid, data-driven evidence supporting the importance of certain metrics for long-term health and physical fitness level, while following good data analysis principles : avoiding data leakage through a participant-level split, choosing evaluation metrics suited to the task, reporting results honestly and discussing the real-world risks of misusing fitness or health predictions.